

E-COMMERCE DATA A

```
# %%
```

```
import pandas as pd
```

```
import matplotlib.pyplot as plt
```

```
import seaborn as sns
```

```
import sqlite3
```

```
import pandas as pd
```

```
# Correct path and encoding
```

```
df = pd.read_csv("../Data/Sample - Superstore.csv.csv", encoding='cp1252')
```

```
print(df.head(10))
```

```
print(df.info())
```

```
print(df.describe())
```

```
print(df.shape)
```

```
print(df.columns)
```

```
# Check for missing values
```

```
print(df.isnull().sum())
```

```
print(df.duplicated().sum())
```

```
print(df.drop_duplicates(inplace=True))
```

```
print(df.shape)
```

```
print(df.columns)
```

```
# KPI
```

```
total_sales = df['Sales'].sum()
```

```
total_profit = df['Profit'].sum()
```

```
total_orders = df['Order ID'].nunique()
```

```
avg_order_value = df['Sales'].mean()
```

```
print("="*40)
```

```
print("  BUISNESS KPI")
print("="*40)
print(f" Total Sales : ${total_sales:,.2f}")
print(f" Total Profit : ${total_profit:,.2f}")
print(f" Total Orders : {total_orders}")
print(f" Average Order Count : {avg_order_value:,.2f}%")
```

```
df['Order Date'] = pd.to_datetime(df['Order Date'])
df['Year'] = df['Order Date'].dt.year
df['Month'] = df['Order Date'].dt.month
df['Ship Date'] = pd.to_datetime(df['Ship Date'])
print(df.head())
```

```
negative_profit = df[df['Profit'] < 0]
print(negative_profit)
Normal_profit = df[df['Profit'] > 0]
print(Normal_profit)
```

```
high_sales = df.sort_values(by='Sales', ascending=False).head(10)
print(high_sales[['Customer Name', 'Product Name','Sales']].head(10))
```

```
low_sales = df.sort_values(by='Sales', ascending=True).head(10)
print(low_sales[['Product Name','Sales']].head(10))
```

```
High_Sale_Regions = df.sort_values(by = 'Sales', ascending=False).head(10)
print(High_Sale_Regions[['Region', 'Country', 'Postal Code','Sales']].head(10))
```

```
low_sale_regions = df.sort_values(by='Sales', ascending=True).head(10)
print(low_sale_regions[['Region','Sales']].head(10))
```

```
df.to_csv("../Data/Sample - Superstore_cleaned.csv", index=False)
```

```
print("Data saved sucessfully")
```

```
conn = sqlite3.connect("../Data/Sample - Superstore.db")
```

```
df.to_sql("Superstore", conn , if_exists="replace", index=False)
```

```
query = ""
```

```
SELECT * FROM Sales LIMIT 5, conn
```

```
""
```

```
print(pd.read_sql_query)
```

```
query = """SELECT Region, Round(SUM(Sales), 2) AS Total_Sales FROM Sales GROUP BY Region  
ORDER BY Total_Sales DESC , conn """
```

```
print(pd.read_sql_query)
```

```
query = """SELECT Category, Round(SUM(Sales), 2) AS Total_Category_Sales FROM Sales  
GROUP BY Category ORDER BY Total_Category_Sales DESC, conn """
```

```
print(pd.read_sql_query)
```

```
query = """SELECT Sub-Category, Round(SUM(Sales), 2) AS Total_Sub_Category_Sales FROM  
Sales GROUP BY Sub-Category ORDER BY Total_Sub_Category_Sales DESC, conn """
```

```
print(pd.read_sql_query)
```

```
query = """SELECT Product Name, Round(SUM(Sales), 2) AS Total_Product_Sales FROM Sales  
GROUP BY Product Name ORDER BY Total_Product_Sales DESC LIMIT 10, conn """
```

```
print(pd.read_sql_query)
```

```
query = """SELECT High_Sales_Regions(Region), Rond(SUM(Sales), 2) AS Total_Sales FROM  
Sales GROUP BY Region ORDER BY Total_Sales DESC LIMIT 10 ,conn """
```

```
print(pd.read_sql_query)
```

```
query = ""
```

```
SELECT Ship Mode , Round(SUM(Sales), 2) AS Total_Ship_Mode FROM Sales GROUP BY Ship  
Mode ORDER BY Total_ship_Mode DESC , conn
```

```
""
```

```
print(pd.read_sql_query)
```

```
plt.style.use('ggplot')
```

```
Region_Sales = df.groupby('Region')['Sales'].sum()
```

```
Region_Sales.plot(kind='bar')
```

```
plt.title('Total Sales by Region')
plt.xlabel('Region')
plt.ylabel('Total Sales')
plt.show()
```

```
Category_Sales = df.groupby('Category')['Sales'].sum()
Category_Sales.plot(kind='bar')
plt.title('Total Sales by Category')
plt.xlabel('Category')
plt.ylabel('Total Sales')
plt.show()
```

```
Category_Sales.plot(kind='pie', autopct='%2.1f%%', figsize=(10,8) )
plt.title('Total Sales by Category')
plt.show()
```

```
Category_profit = df.groupby('Category')['Profit'].sum().sort_values(ascending=False)
Category_profit.plot(kind='pie')
plt.title('Category Profit')
plt.xlabel('CATEGORY')
plt.ylabel('PROFIT')
plt.show()
```

```
Sub_Category_profit = df.groupby('Sub-Category')['Sales'].sum().sort_values(ascending=False)
Sub_Category_profit.plot(kind='pie')
plt.title('Sub - Category Profit')
plt.xlabel("")
plt.ylabel("")
plt.show()
```

```
Sub_Category_Sales = df.groupby('Sub-Category')['Sales'].sum()
```

```
Sub_Category_Sales.plot(kind='bar')

plt.title('Total Sales by Sub-Category')

plt.xlabel('Sub-Category')

plt.ylabel('Total Sales')

plt.show()
```

```
Product_Name_Sales = df.groupby('Product
Name')['Sales'].sum().sort_values(ascending=False).head(10)

Product_Name_Sales.plot(kind='bar')

plt.title('Top 10 Product Sales')

plt.xlabel('Product Name')

plt.ylabel('Total Sales')

plt.xticks(rotation=45, ha='right')

plt.show()
```

```
High_Sales_Regions =
df.groupby('Region')['Sales'].sum().sort_values(ascending=False).head(10)

High_Sales_Regions.plot(kind='bar')

plt.title('Top 10 Sales by Region')

plt.xlabel('Region')

plt.ylabel('Total Sales')

plt.show()
```

```
top_customer = df.groupby('Customer
Name')['Profit'].sum().sort_values(ascending=False).head(10)

top_customer.plot(kind='pie')

plt.title('Top Customer')

plt.xlabel("")

plt.ylabel("")

plt.show()
```

```
discount_profit = df.groupby('Discount')['Profit'].sum().sort_values(ascending=False).head(10)
```

```
discount_profit.plot(kind='bar')  
  
plt.title('Discount profit')  
  
plt.xlabel("")  
  
plt.ylabel("")  
  
plt.show()
```

```
monthly_sales = df.groupby(['Year' , 'Month'])['Sales'].sum()  
  
monthly_sales.plot(kind='bar')  
  
plt.title('Monthly Sales')  
  
plt.xlabel("")  
  
plt.ylabel("")  
  
plt.show()
```

```
monthly_profit = df.groupby(['Year' , 'Month'])['Profit'].sum()  
  
monthly_sales.plot(kind='pie')  
  
plt.title('Monthly Profit')  
  
plt.xlabel("")  
  
plt.ylabel("")  
  
plt.show()
```

```
state_profit = df.groupby('City')['Profit'].sum().sort_values(ascending=False)  
  
state_profit.plot(kind='bar')  
  
plt.title('State Profit')  
  
plt.xlabel("")  
  
plt.ylabel("")  
  
plt.show()
```

```
ship_profit = df.groupby('Ship Mode')['Profit'].sum().sort_values(ascending=False)  
  
ship_profit.plot(kind='line')  
  
plt.title('Shipwise profit')  
  
plt.xlabel("")
```

```
plt.ylabel("")
```

```
plt.show()
```

```
correlation = df[['Sales', 'Profit', 'Quantity', 'Discount']].corr()
```

```
print(correlation)
```

```
plt.figure(figsize=(8, 6))
```

```
sns.heatmap(correlation, annot=True, cmap='coolwarm')
```

```
plt.title('Correlation Heatmap')
```

```
plt.show()
```

```
# visulaization
```

```
Region_Sales.plot(kind='pie', autopct='%2.1f%%', figsize=(8, 8))
```

```
plt.title('Region Distribution')
```

```
plt.xlabel("")
```

```
plt.ylabel("")
```

```
plt.show()
```

```
Product_Name_Sales.plot(kind='pie', autopct='%1.1f%%', figsize=(8, 8))
```

```
plt.title("")
```

```
plt.xlabel('Product Name')
```

```
plt.ylabel('Total Sales')
```

```
plt.xticks(rotation=45, ha='right')
```

```
plt.show()
```

```
Category_Sales.plot(kind='pie', autopct='%1.1f%%', figsize=(8,8) )
```

```
plt.title('Total Sales by Category')
```

```
plt.show()
```

```
Sub_Category_Sales = df.groupby('Sub-Category')['Sales'].sum()
```

```
Sub_Category_Sales.plot(kind='pie', autopct='%2.1f%%', figsize=(10,10))
```

```
plt.title('Total Sales by Sub-Category')
```

```
plt.show()
```

```
print("Highest sales region is WEST with more than 70K")
```

```
print("Highest sales category is TECHNOLOGY")
```

```
print("Technology generates highest profit")
```

```
print("Canon images are very profitable and highly sold")
```

```
print("TAMARA CHAND is the top customer")
```

```
print("Monthly sale is highest in NOVEMBER OF 2017")
```

```
print("STANDARD CLASS SHIP mode generates high profit")
```

```
print("AT 0.0 DISCOUNT profit is 30k")
```

```
print("In Sub-Category phones and chairs generates highest profit")
```

```
print("In subcategory FASTENERS are loss making")
```

```
print("Keith Dawkins is uninterested customer")
```

```
print("Monthly sale is very less in March, 2013")
```